

Supporting Information

Reduction Kinetics of the three Nitrothiophenol Isomers on Pt-coated Gold-Nanorods

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Pt@AuNR characterization.

UV/Vis Absorption Spectra of Gold Cluster:

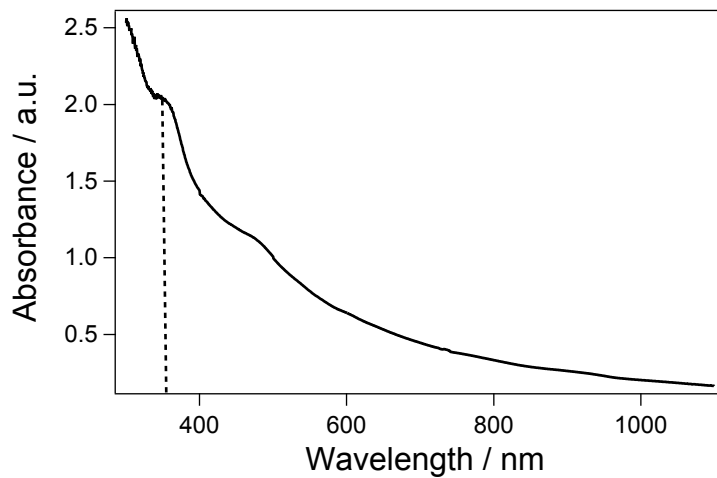


Figure S1. UV/Vis absorption spectrum of the synthesized gold clusters (OD = 2.1, at 360 nm).

TEM Image and UV/Vis Absorption Spectra of Small Gold Nanorods (AuNR, with 26 nm in length and 13 nm in width):

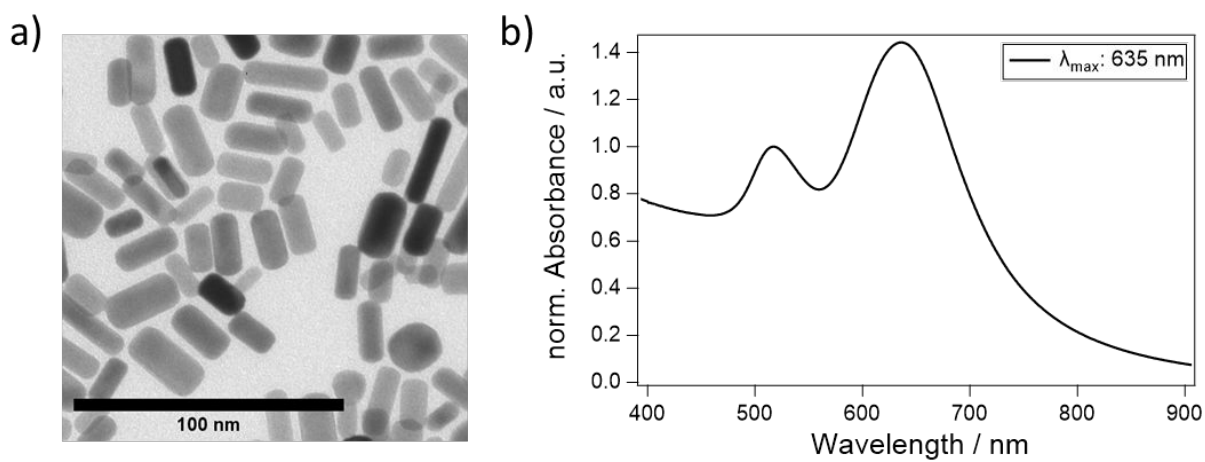


Figure S2 a) TEM image of the synthesized AuNR (L: 26 ± 4 nm, B: 13 ± 2 nm) measured at with 120 kV accelerating voltage (Joel JEM 1400 Plus) and **b)** the UV/Vis absorption spectrum ($\lambda_{\text{max}} = 635$ nm) measured with a UV/Vis absorption spectrophotometer of the type V-730 from JASCO.

TEM Image and UV/Vis Absorption Spectra of Gold Nanorods (AuNR with 52 nm in length and 23 nm in width):

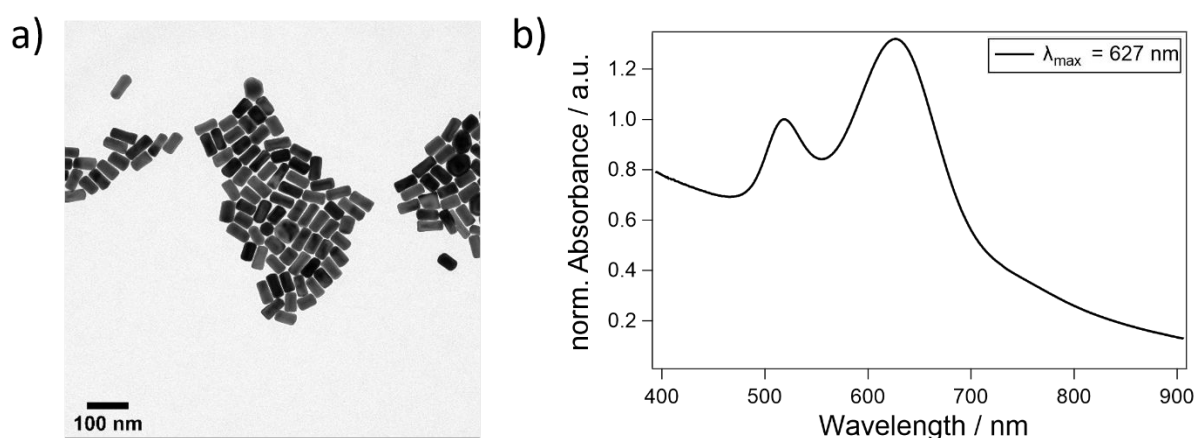


Figure S3 a) TEM image of the synthesized AuNR (L: 52 ± 6 nm, B: 23 ± 3 nm) measured with 120 kV accelerating voltage (Joel JEM 1400 Plus) and **b)** the UV/Vis absorption spectrum ($\lambda_{\text{max}} = 627$ nm) measured with a UV/Vis absorption spectrophotometer of the type V-730 from JASCO.

TEM Image and UV/Vis Absorption Spectra Synthesis of Platinated Gold Nanorods (Pt@ AuNR with 55 nm in length and 24 nm in width):

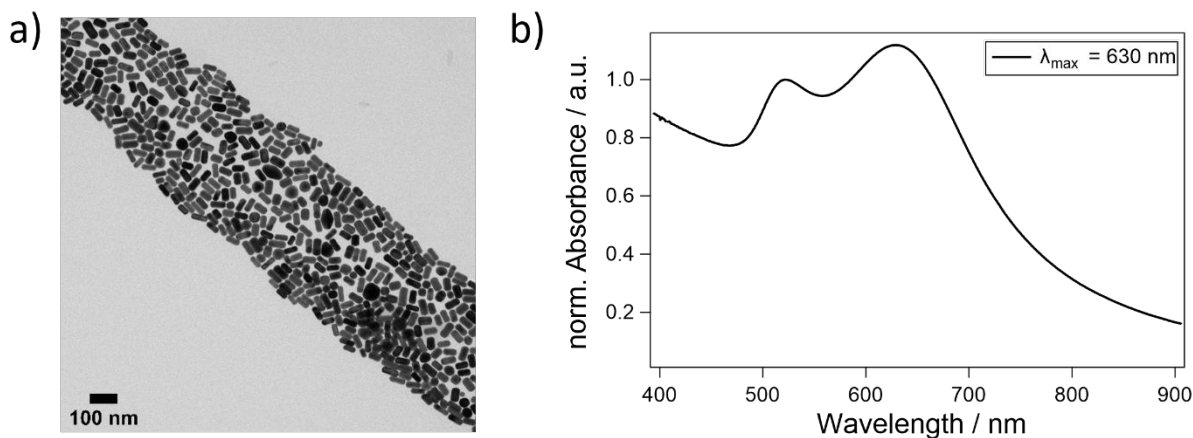


Figure S4 a) TEM image of the synthesized Pt@AuNR (L: $55 \pm 5 \text{ nm}$, B: $24 \pm 3 \text{ nm}$) measured with 120 kV accelerating voltage (Joel JEM 1400 Plus) and b) the UV/Vis absorption spectrum ($\lambda_{\text{max}} = 630 \text{ nm}$) measured with a UV/Vis absorption spectrophotometer of the type V-730 from JASCO.

AuNR characterization and FDTD Simulation

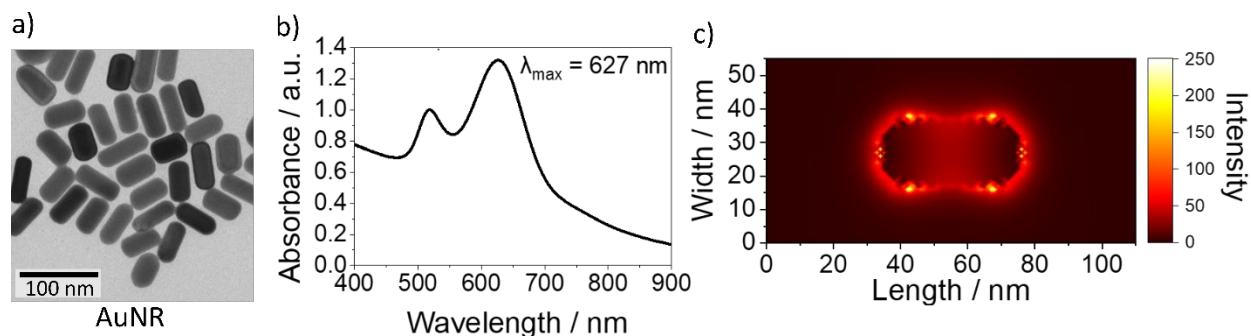


Figure S5 a) TEM image of the used AuNR (L: $52 \pm 6 \text{ nm}$, W: $23 \pm 3 \text{ nm}$) and b) corresponding UV/Vis absorption spectra ($\lambda_{\text{max}} = 627 \text{ nm}$) to the nanorods shown in a); c) FDTD simulation for the AuNR to determine the theoretical electric field enhancement based on the sizes of the synthesized nanostructures.

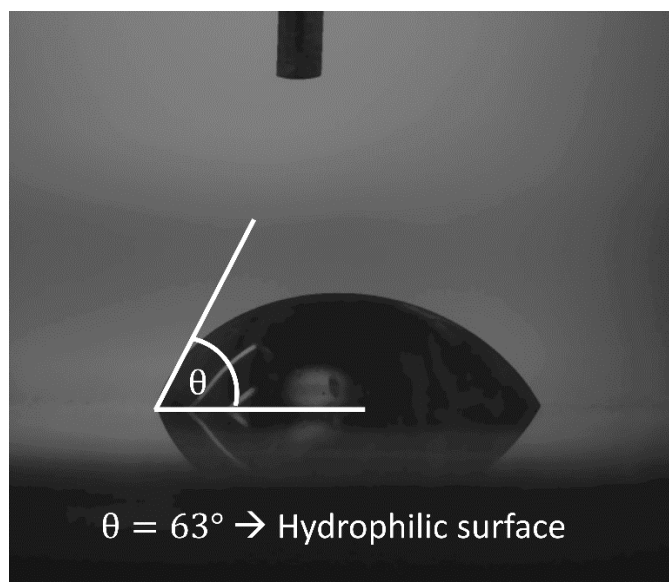


Figure S6 shows the result of the contact angle measurement, which was performed with the molecularly modified nanoparticles deposited on a glass slide. The contact angle θ is 63° and thus describes the surface as hydrophilic.

Raman spectra of the three NTP-Isomers

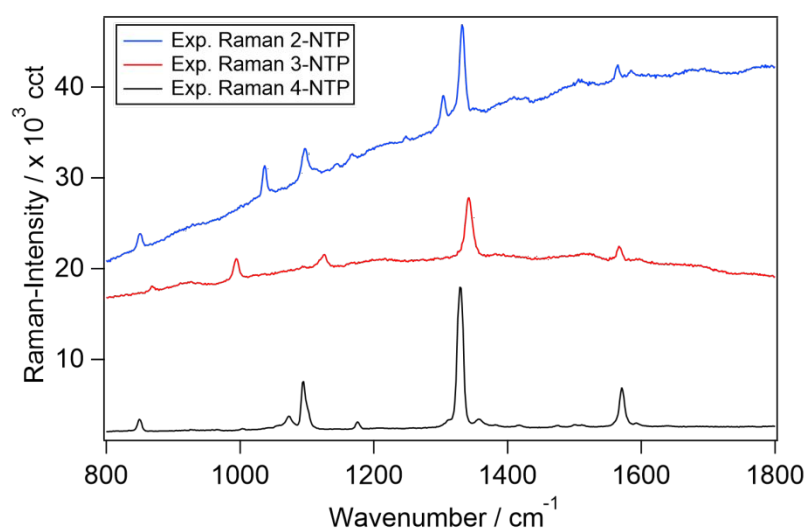


Figure S7: Experimental raw Raman spectra of the three NTP isomers, 2-NTP (blue), 3-NTP (red), 4-NTP (black). The spectra were obtained using a 633 nm HeNe-Laser, 497 μ W laser power, an integration time of 30 s and three accumulations.

Density functional theory calculations

Density functional theory calculations were used to assign the vibrational bands of the nitrothiophenol reactants and the aniline derivative products.

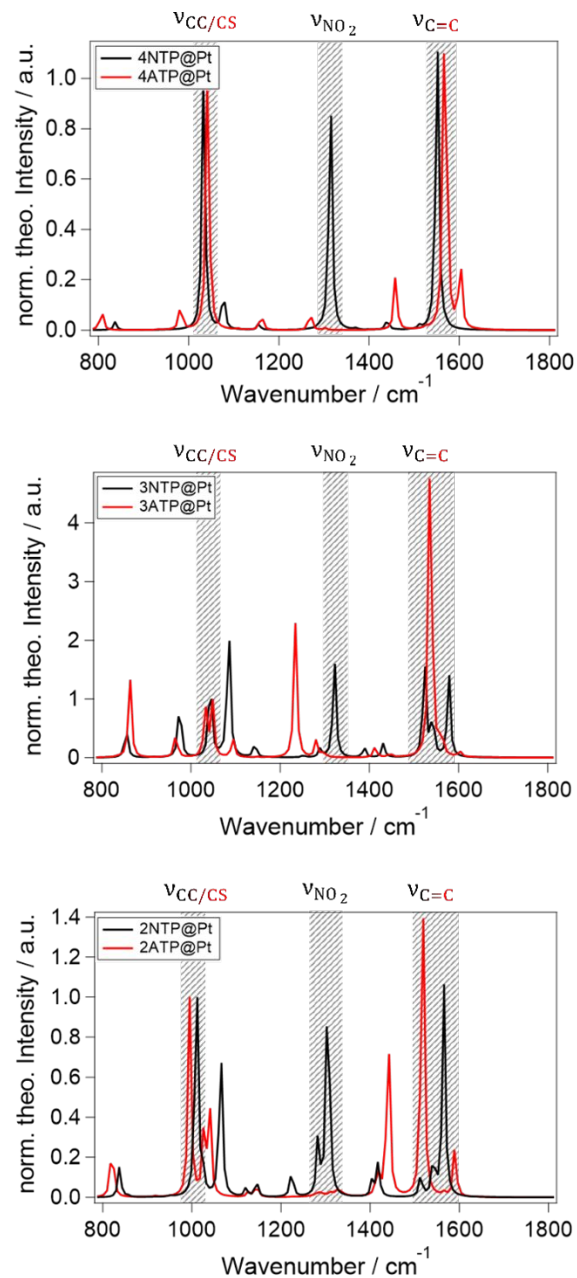


Figure S8: Calculated Raman spectra of the nitrothiophenol educts and the corresponding products, the aminothiophenols bound to a platinum atom in a wavenumber range of 800-1800 cm^{-1} . The spectra were calculated with the B3LYP 6-31+G** basis set.

Nitrothiophenol isomer reduction monitoring

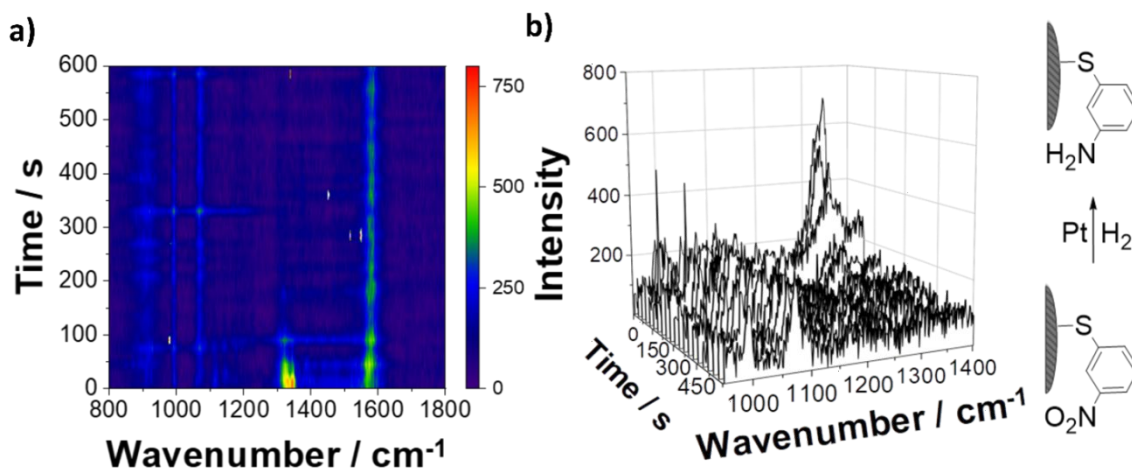


Figure S9: shows in **a)** a contour plot for the 3-NTP reduction on the Pt@AuNR during a reaction time up to 600 seconds. and **b)** the recorded raw data for the reduction in a wavenumber range of 950-1400 cm^{-1} to better highlight the spectral differences. The spectra were obtained using a 633 nm HeNe-Laser, 5 mW laser power, an integration time of 5 s and three accumulations.

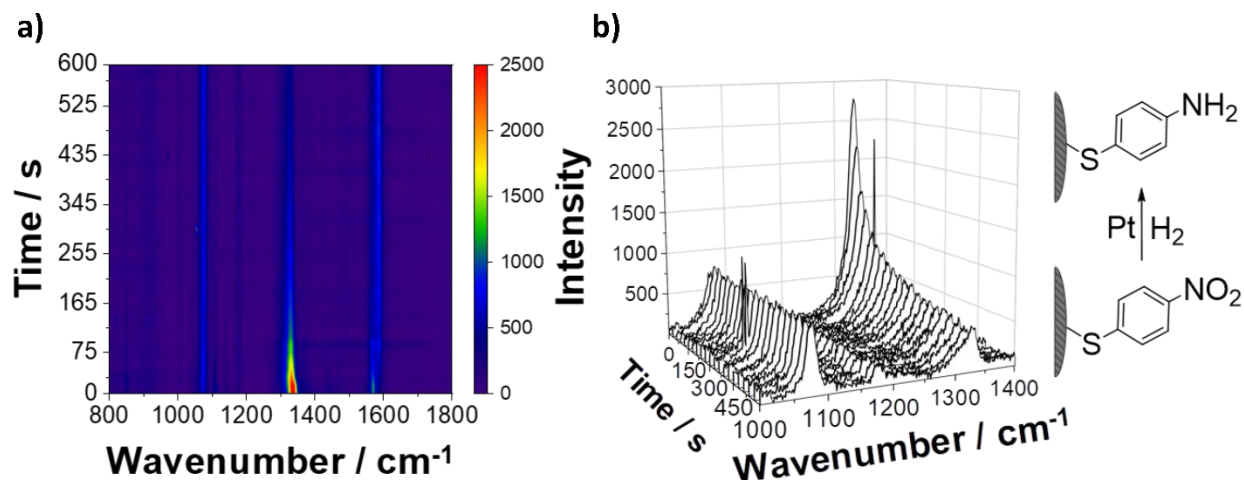


Figure S10: shows **a)** a contour plot for the 4-NTP reduction on the Pt@AuNR during a reaction time up to 600 seconds. and **b)** the recorded raw data for the reduction in a wavenumber range of 1000-1400 cm^{-1} to better highlight the spectral differences. The spectra were obtained using a 633 nm HeNe-Laser, 5 mW laser power, an integration time of 5 s, and three accumulations.

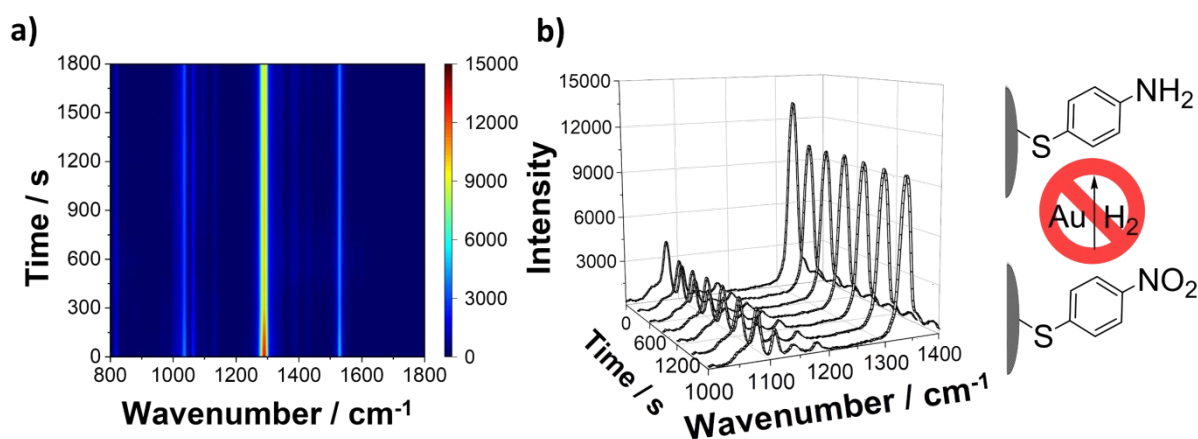


Figure S11: shows **a)** a contour plot for the 4-NTP reduction with AuNR during a reaction time up to 600 seconds. and **b)** the recorded raw data for the reduction in a wavenumber range of 1000-1400 cm^{-1} to better highlight the spectral differences. The spectra were obtained using a 633 nm HeNe-Laser, 5 mW laser power, an integration time of 5 s, and three accumulations.

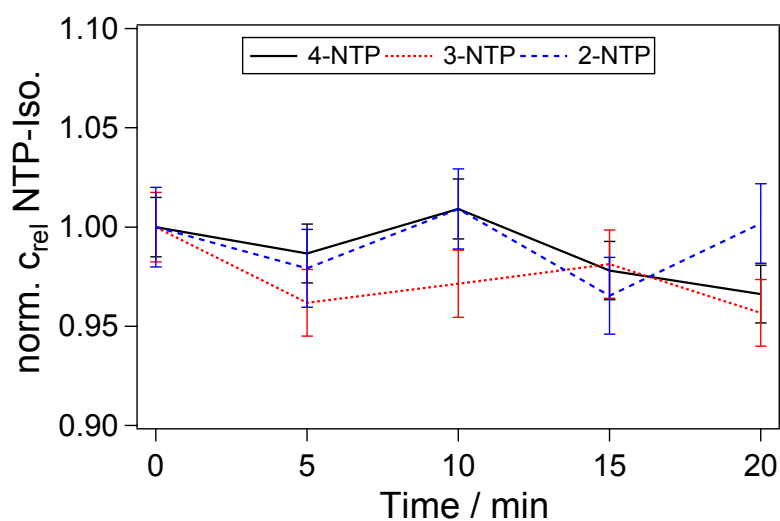


Figure S12: Control plot of the SERS monitoring of the catalytic reduction of the NTP-isomers (4-NTP = black; 3-NTP = red; 2-NTP= blue) using Pt@AuNR as the catalyst and no hydrogen for a time up to 20 min. Norm. C_{rel} stands for the normalized relative concentration of the NTP-isomers.

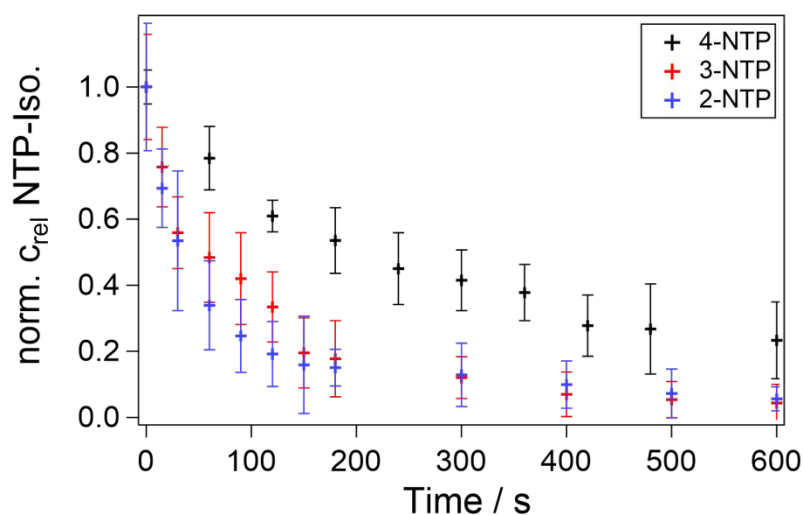


Figure S13: Zero-order kinetic-plot of the SERS monitoring of the catalytic reduction of the NTP-isomers (4-NTP = black; 3-NTP = red; 2-NTP= blue) using Pt@AuNR as the catalyst and

hydrogen as reducing agent for a reduction time up to 600 s. C_{rel} stands for the relative concentration of the NTP-isomers.

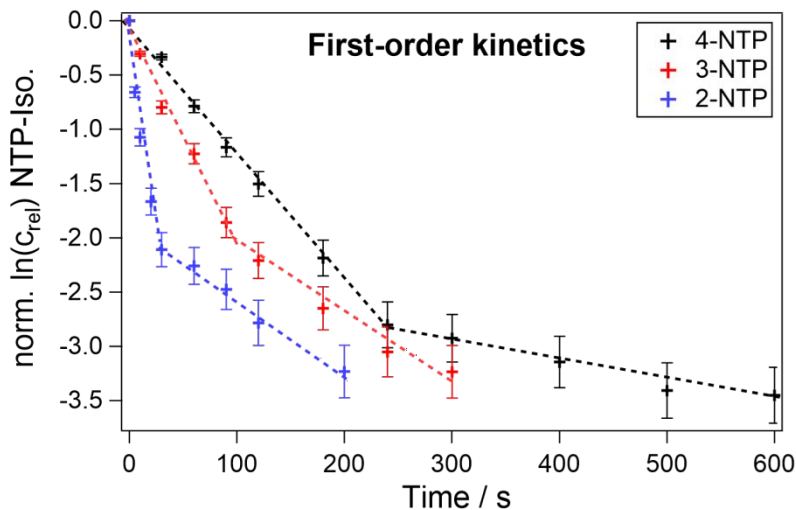


Figure S14: 1. Order kinetic-plot of the SERS monitoring of the NTP-isomers (4-NTP = black; 3-NTP = red; 2-NTP= blue) using Pt@AuNR as the catalyst and sodium borohydride (NaBH_4) as reducing agent for a reaction time up to 600 s. The reductions follow first-order kinetics for all three isomers, showing a decrease in the reduction rate after an inflection point. C_{rel} stands for the relative calculated concentration of the NTP-isomers.

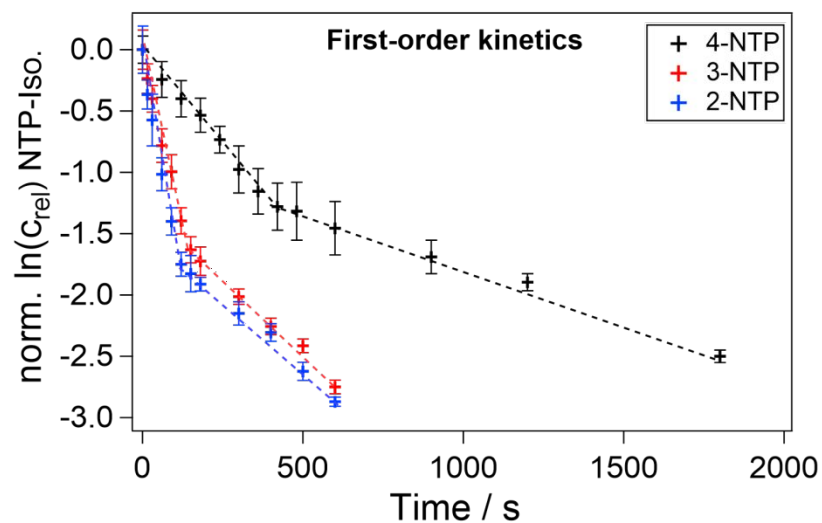


Figure S15: First-order kinetic-plot of SERS monitoring of the NTP-isomers (4-NTP = black; 3-NTP = red; 2-NTP= blue) using Pt@AuNR as the catalyst and hydrogen as the reducing agent for the whole reduction time (1800s). The reductions follow first-order kinetics for all three isomers, showing a decrease in the reduction rate after an inflection point. C_{rel} stands for the relative concentration of the NTP-isomers.